American Computer Science League

2025-2026 ● Contest 2: License Plates ● Junior Division

PROBLEM: Cars in the People's Republic of China have 2 kinds license plate numbers, each with its own rules and restrictions:

- green plates for energy-efficient cars
  - starts with a letter and a dot
  - contains 6 alphanumeric characters after that

- blue plates for gas-powered cars
  - starts with a letter and a dot
  - contains 5 alphanumeric characters after that
  - the last alphanumeric character is a digit that restricts cars from being driven on certain days

All alphabetic characters are uppercase. In this program, if the last digit of a blue plate is odd, it is "restricted" from being driven on Sunday, Tuesday, Thursday, and Saturday. A license plate that has adjacent digits that are repeated 6s, repeated 8s, or repeated 9s is considered "lucky". ACSL has added that it is a "fortunate" plate number if there are either at least 2 consecutive, adjacent letters or at least 3 consecutive, adjacent digits in strictly ascending or strictly descending order among those that are after the dot.

You will be given a license plate number and a day of the week. For a valid plate, output one of the following strings in the following priority order: "fortunate", "lucky", "restricted", or the plate color. Otherwise output "invalid".

EXAMPLE:

Input	`F.11235` `Thursday`
Output	`fortunate`
Explanation	This license plate is a valid blue license plate because the first character is a letter followed by a dot and there are 5 alphanumeric characters following with the last one being a digit. It is "restricted" since the last digit is odd and it's Thursday. It has no repeated sequence of 6s, 8s, or 9s, so it is not "lucky". It is "fortunate" because 123 are 3 consecutive, adjacent digits. Based on the specified priority order, output the string `fortunate.`

INPUT: There will be two strings representing a license plate number and a day of the week.

OUTPUT: Output one of 6 possible strings: “blue”, “green”, “lucky”, “restricted”, "fortunate", or “invalid”.

SAMPLE INPUT:

```
1. F.112358
   Thursday
2. C.ARLEN
   Friday
3. A.M1437
   Saturday
4. D.RRC888
   Sunday
5. P.062825
   Monday
6. Z.47786
   Tuesday
```

SAMPLE OUTPUT:

1. fortunate
2. invalid
3. restricted
4. lucky
5. green
6. blue

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TEST DATA

TEST INPUT:

```
1. X.44997
   Saturday
2. P.31415
   Thursday
3. E.F2171
   Friday
4. KELVIN0
   Saturday
5. P.543210
   Sunday
6. J.CABNYC
   Monday
```

TEST OUTPUT:

```
1. lucky
2. restricted
3. blue
4. invalid
5. fortunate
6. fortunate
```

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问题: 中华人民共和国的汽车主要使用两种类型的车牌号码，每种都有各自的规则和限制：

- 新能源汽车 - 绿色（green）车牌：
  - 以**一个字母和一个点**号**开头**
  - 其后紧跟 6 个字母数字字符

- 燃油汽车 - 蓝色（blue）车牌：
  - 以**一个字母和一个点**号**开头**
  - 其后紧跟 5 个字母数字字符
  - 最后一位数字字符决定了车辆在特定日期是否限行（restricted）

车牌中的所有字母均为大写。在本程序中，**如果蓝色车牌的最后一位数字是奇数，则该车在** Sunday（**周日**）、Tuesday（周二）、Thursday（**周四**）**和** Saturday（**周六**）**为“** r-estricted（限行）**”状态。如果车牌中包含相邻且重复的数字** 6、8 **或** 9，**则该车牌被视为“** lucky（**幸运**）**”车牌。**ACSL **补充规定，如果在点号之后的字符中，满足以下任一条件，则为“** fortunate（吉利）**”车牌：存在至少** 2 **个连续相邻的字母；存在至少** 3 **个连续相邻的数字，且为严格升序或严格降序排列。**

给到一个车牌号码和一个代表星期几的单词。**如果车牌有效，请按照以下优先级顺序输出其中一个字符串：**“ fortunate（吉利）”, “ lucky（幸运）”, “ restricted（限行）”或车牌颜色（blue/green）。**如果车牌无效，则输出** invalid。

示例:

输入	`F.11235` `Thursday`
输出	`fortunate`
解析	这是一个有效的蓝色车牌，因为它以字母开头，**紧跟**一个点号，且后面有 5 个字母数字字符，其中最后一个字符是数字。**由于最后一位数字**字符**是奇数，且输入日期是** Thursday，**所以它是** “ restricted ”状态。**因为它没有重复的** 6、8 **或** 9 **序列**，所以**它不是“** lucky **”车牌。**又**因为字符中包含** 123 **这** 3 **个连续**且**相邻**的数字，所以**它**是**“** fortunate **”车牌。根据给定的优先级顺序，最终输出字符串** `fortunate`。

输入: 输入两个字符串，**它们**分别表示车牌号码和**一个**代表星期几的单词。

输出: 输出应为以下六个字符串之一："blue"、"green"、"lucky"、"restricted"、"fortunate"或"invalid"。

样例输入:

1. `F.112358`
   `Thursday`
2. `C.ARLEN`
   `Friday`
3. `A.M1437`
   `Saturday`
4. `D.RRC888`
   `Sunday`
5. `P.062825`
   `Monday`
6. `Z.47786`
   `Tuesday`

样例输出:

1. `fortunate`
2. `invalid`
3. `restricted`
4. `lucky`
5. `green`
6. `blue`

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测试数据

测试输入：

1. `X.44997`
   `Saturday`
2. `P.31415`
   `Thursday`
3. `E.F2171`
   `Friday`
4. `KELVIN0`
   `Saturday`
5. `P.543210`
   `Sunday`
6. `J.CABNYC`
   `Monday`

测试输出：

1. `lucky`
2. `restricted`
3. `blue`
4. `invalid`
5. `fortunate`
6. `fortunate`